Thermodynamic Properties of Real Gases from Cubic Equations of State

PVT behaviour, residual enthalpy and entropy, and pure-component fugacity

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1 Scope

The program `cubic_eos.py` (Python 3, NumPy/SciPy) covers the three stages of the project for a pure gas: (i) PVT behaviour from four cubic equations of state (EOS) – van der Waals (vdW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR); (ii) the thermodynamic properties H and S (residual and total); (iii) the pure-component fugacity f and fugacity coefficient ϕ, including the vapour pressure from equal fugacity. A database of 22 gases (T_c, P_c, ω, ideal-gas C_p) is included and any other gas can be entered by hand.

2 Theory

Generic cubic EOS. All four models are special cases of

$$P = \frac{RT}{V-b} - \frac{a(T)}{(V+\epsilon b)(V+\sigma b)}, \qquad b = \Omega\frac{RT_c}{P_c}, \quad a(T) = \Psi\,\alpha(T_r)\frac{R^2T_c^2}{P_c}, \tag{1}$$

with the constants of Table 1 ($\kappa_{SRK} = 0.480 + 1.574\omega - 0.176\omega^2$, $\kappa_{PR} = 0.37464 + 1.54226\omega - 0.26992\omega^2$, $\alpha = [1+\kappa(1-\sqrt{T_r})]^2$).

EOS	σ	ϵ	Ω	Ψ	$\alpha(T_r)$
vdW	0	0	1/8	27/64	1
RK	1	0	0.08664	0.42748	$T_r^{-1/2}$
SRK	1	0	0.08664	0.42748	$[1+\kappa_{SRK}(1-\sqrt{T_r})]^2$
PR	$1+\sqrt{2}$	$1-\sqrt{2}$	0.07779	0.45724	$[1+\kappa_{PR}(1-\sqrt{T_r})]^2$

Table 1: Constants of the generic cubic EOS (the program uses the unrounded values).

PVT. With $A = aP/(RT)^2 = \Psi\alpha P_r/T_r^2$ and $B = bP/RT = \Omega P_r/T_r$, Eq. (1) becomes the cubic in $Z = PV/RT$

$$Z^3 + \big[(\epsilon+\sigma)B - (1+B)\big]Z^2 + \big[\epsilon\sigma B^2 - (1+B)(\epsilon+\sigma)B + A\big]Z - \big[(1+B)\epsilon\sigma B^2 + AB\big] = 0. \tag{2}$$

Three real roots occur below T_c in a pressure window (the van der Waals loop); the smallest is liquid-like, the largest vapour-like, the middle one is mechanically unstable. For (T, V) input, P is evaluated directly from Eq. (1).

Fugacity. With $q = A/B$ and $I = \frac{1}{\sigma-\epsilon}\ln\frac{Z+\sigma B}{Z+\epsilon B}$ ($I = B/Z$ for vdW),

$$\ln\phi = Z - 1 - \ln(Z-B) - qI, \qquad f = \phi P. \tag{3}$$

When three roots exist the stable phase is the one with the lowest $\ln\phi$ (lowest Gibbs energy). The vapour pressure $P^{sat}(T)$ is the pressure at which $\ln\phi^L = \ln\phi^V$.

Enthalpy and entropy. The residual (departure) functions follow from $H^R/RT = -T(\partial\ln\phi/\partial T)_P$ and $S^R/R = H^R/RT - \ln\phi$; with $d\ln\alpha/d\ln T_r$ (0, $-\frac{1}{2}$, $-\kappa\sqrt{T_r}/\sqrt{\alpha}$ for vdW, RK, SRK/PR):

$$\frac{H^R}{RT} = Z - 1 + \Big(\frac{d\ln\alpha}{d\ln T_r} - 1\Big)qI, \qquad \frac{S^R}{R} = \ln(Z-B) + \frac{d\ln\alpha}{d\ln T_r}\,qI, \qquad G^R = RT\ln\phi = H^R - TS^R. \tag{4}$$

The total properties add the ideal-gas part, with $C_p^{ig}/R = A_0 + B_0T + C_0T^2 + D_0T^{-2}$ and reference $T_0 = 298.15$ K, $P_0 = 1$ bar ($H^{ig} = S^{ig} = 0$):

$$H = \int_{T_0}^{T} C_p^{ig}dT + H^R, \qquad S = \int_{T_0}^{T}\frac{C_p^{ig}}{T}dT - R\ln\frac{P}{P_0} + S^R. \tag{5}$$

Property *changes* between two states, $\Delta H = H_2 - H_1$ and $\Delta S = S_2 - S_1$, are independent of the reference state; the program calculates them for any pair $(T_1, P_1) \to (T_2, P_2)$, including across a phase change.

3 Program structure and algorithm

1. **Input (prompted):** gas (name/formula or custom T_c, P_c, ω, C_p coefficients), EOS, and (T, P) or (T, V); units such as `25C`, `5MPa`, `1atm`, `700psi` are accepted.
2. **Roots:** the cubic (2) is solved with `numpy.roots`; each root is polished by Newton iteration (important for the tiny liquid root at low P) and roots with $Z \le B$ ($V \le b$) are rejected.
3. **Phase and properties:** for every root Z, $\ln\phi$, H^R, S^R, H, S are evaluated by Eqs. (3)–(5). The root with minimum $\ln\phi$ is the stable phase; the label (vapour, liquid, supercritical fluid, saturated) is obtained by comparison with $P^{sat}(T)$ and T_c.

4. **Saturation:** the two extrema of the isotherm are located from $\partial P/\partial V = 0$ (a quartic in V); they bracket P^{sat}, which is found by Brent's method on $g(\ln P) = \ln \phi^L - \ln \phi^V$ (it returns $\Delta H^{vap} = H^V - H^L$ and $\Delta S^{vap} = \Delta H^{vap}/T$).
5. **Interface:** running `python cubic_eos.py` starts a guided dialogue that prompts for everything needed (gas, EOS, type of calculation, T, P or V) and repeats until the user quits. Menu: state at (T, P); state at (T, V); vapour pressure, saturated fugacity and ΔH^{vap}; table versus P with optional CSV export; $\Delta H, \Delta S$ between two states; comparison of all four EOS.

Robustness: the solver was run for all 22 gases × 4 EOS at $T_r = 0.45, 0.5, 0.6, 0.8, 0.95, 0.99, 0.999$ (644 cases) without failure.

4 Verification

`validate.py` runs the following automatic tests (all pass).

Test	Result
Ideal-gas limit ($P \to 0$) for all gases/EOS	$Z, \phi \to 1$, $H^R, S^R \to 0$
$H^R = -RT^2(\partial \ln \phi/\partial T)_P$ and S^R identity, 828 states (central differences)	max. error 9×10^{-8} in H^R/RT and S^R/R
Equal fugacity at P^{sat} (112 cases)	$\lvert \ln \phi^L - \ln \phi^V \rvert < 10^{-14}$
Clapeyron: ΔH^{vap} from H^R vs. $T\Delta V\, dP^{sat}/dT$	relative difference $\leq 1.5 \times 10^{-7}$
Closed-form $\int C_p\, dT$, $\int C_p/T\, dT$ vs. numerical quadrature	3×10^{-16}
Independent code (CoolProp PR and SRK backends, 58 states)	Z, $\ln \phi$, H^R, S^R agree to $< 10^{-8}$

Table 2: Verification tests. The last row confirms Eqs. (2)–(4) are implemented exactly as in an independent code.

5 Results

PVT and phase behaviour. Fig. 1a shows PR isotherms of n-butane with the van der Waals loop below T_c and the saturation dome from the fugacity calculation; the critical isotherm has the inflection at P_c. Fig. 1b compares $Z(P)$ of CO_2 at 350 K ($T_r = 1.15$) with the CoolProp reference Helmholtz EOS: all models are exact in the limit $P \to 0$; vdW is poor, PR and RK follow the reference, SRK over-predicts Z around 150–250 bar.

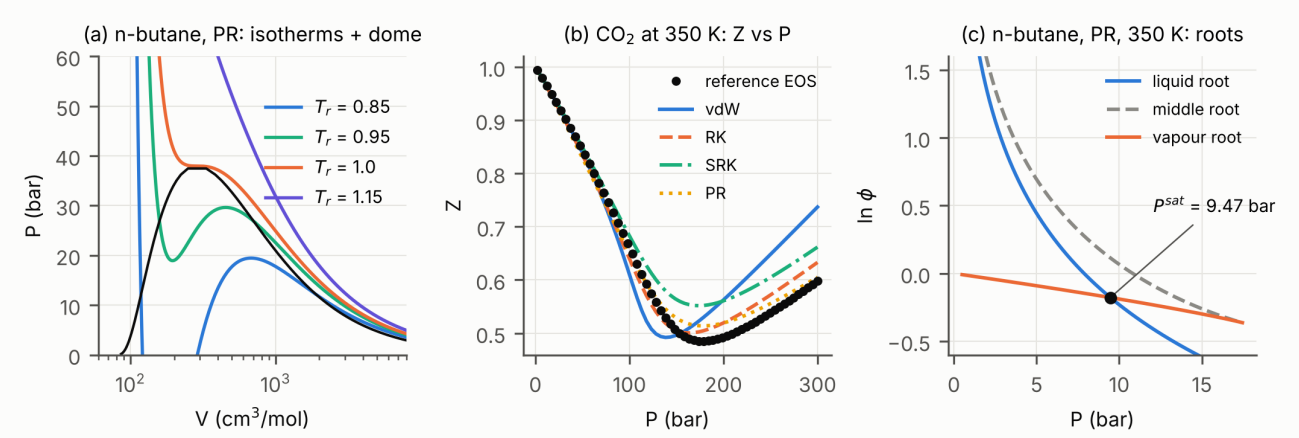


Figure 1: (a) PR isotherms and saturation dome of n-butane. (b) Z of CO_2 at 350 K, four EOS vs. reference EOS. (c) $\ln \phi$ of the three roots of the cubic for n-butane (PR, 350 K): the stable root is the lower branch, and the liquid and vapour branches cross at P^{sat} (equal fugacity).

Fugacity. Fig. 1c illustrates the stability rule: between the two spinodal pressures all three roots exist and the stable one is the lowest in $\ln \phi$; at $P^{sat} = 9.47$ bar (experimental/reference 9.44 bar) the curves cross, $f^{sat} = 7.93$ bar, $\phi^{sat} = 0.837$. Fig. 2a gives ϕ of CO_2 at 350 K, which falls to 0.44 at 300 bar, i.e. the real gas has a much lower escaping tendency than an ideal gas at the same P.

Enthalpy and entropy. Fig. 2b,c show that attractive forces make H^R and S^R increasingly negative with P; vdW and RK underestimate the magnitude at high P whereas SRK and PR follow the reference closely. Table 3 lists a worked state; Table 4 summarises the accuracy over a $T_r = 1.1$–2, $P_r = 0.2$–4 grid (20 states per gas, five gases: CO_2, CH_4, C_3H_8, N_2, NH_3) against the reference EOS.

Saturation and ΔH^{vap}. For n-butane at 350 K the program gives $P^{sat} = 16.60$ (vdW), 11.41 (RK), 9.59 (SRK), 9.47 (PR) bar against 9.44 bar from the reference, and $\Delta H^{vap} = 8.17$, 14.69, 17.55, 17.47 kJ/mol against 17.30 kJ/mol. Only SRK and PR, whose $\alpha(T_r)$ is fitted to vapour pressure through ω, are accurate; the same holds for

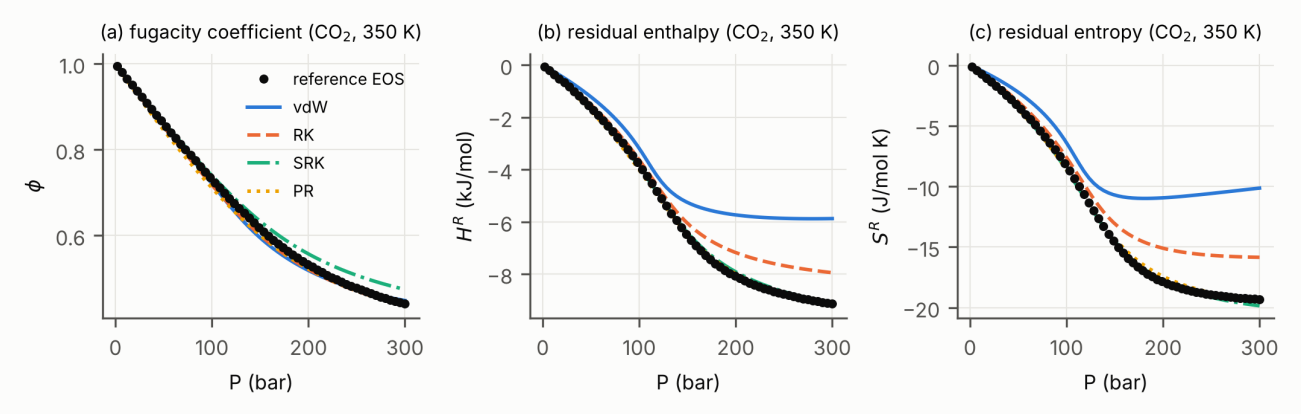


Figure 2: Fugacity coefficient, residual enthalpy and residual entropy of CO_2 at 350 K versus pressure, four EOS and reference EOS.

CO_2, 350 K, 100 bar	Z	ϕ	f (bar)	H^R (J/mol)	S^R (J/mol K)
vdW	0.6106	0.721	72.1	−3190	−6.40
RK	0.6400	0.718	71.8	−3628	−7.61
SRK	0.6831	0.740	74.0	−3900	−8.64
PR	0.6511	0.711	71.1	−4014	−8.63
Reference EOS	0.6610	0.731	73.1	−3830	−8.34

Table 3: Worked example (stable phase: supercritical-temperature gas).

propane, CO_2 and NH_3 (e.g. NH_3 at 324.6 K: PR 20.97 bar, reference 21.08 bar). **Property change:** compressing N_2 from 300 K, 1 bar to 400 K, 100 bar (PR) gives $\Delta H = 2633$ J/mol and $\Delta S = -30.77$ J/(mol K), versus 2927 J/mol and −29.87 J/(mol K) for an ideal gas, so neglecting the residual terms would be a 10% error in ΔH.

6 Discussion and conclusions

- The unified $(\sigma, \epsilon, \Omega, \Psi, \alpha)$ formulation lets one set of formulae (Eqs. 2–5) serve all four EOS; adding a further cubic EOS needs only one new row and its $\alpha(T_r)$ derivative.
- Thermodynamic consistency is demonstrated numerically: H^R, S^R obey the exact relations with $\ln\phi$, fugacities are equal at P^{sat} and ΔH^{vap} satisfies Clapeyron's equation.
- Accuracy: PR gives the best Z and P^{sat} overall, SRK the best H^R, S^R in this gas set; vdW is only qualitative. Liquid volumes and the near-critical region are only approximate (PR gives V^L 2% low for n-butane at 350 K; no volume translation was used), and for strongly polar fluids (NH_3, water) errors are larger. In water, PR predicts a vapour pressure of 0.96 bar at 100°C instead of 1.013 bar.
- Limitations: pure components only; ideal-gas C_p polynomials are valid for 298–1500 K; temperatures far below $T_r \approx 0.45$ (extremely small P^{sat}) are not guaranteed.

Files submitted: `cubic_eos.py` (program), `validate.py` (tests), `make_figures.py` (figures and example data), this report. **Data/references:** Smith, Van Ness & Abbott, *Introduction to Chemical Engineering Thermodynamics*, Tables B.1 and C.1 and Ch. 3, 6, 10; Peng & Robinson, Ind. Eng. Chem. Fundam. 15 (1976) 59; Soave, Chem. Eng. Sci. 27 (1972) 1197; Bell et al., CoolProp, Ind. Eng. Chem. Res. 53 (2014) 2498 (reference values used for comparison only).

Mean absolute error vs. reference (%)	vdW	RK	SRK	PR
Z	5.2	2.1	3.3	2.0
H^R	14.9	4.7	3.6	9.2
S^R	21.1	7.0	5.7	8.0

Table 4: Average accuracy of the four EOS (100 states).